

University of Dhaka
Affiliated Engineering Colleges
Department of Computer Science and Engineering
1st Year 1st Semester B.Sc. in CSE Final Examination, 2021
CSE-1102: Discrete Mathematics

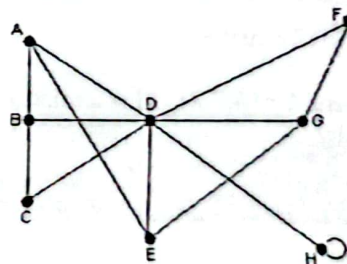
Total Marks: 70

Time: 3 Hours

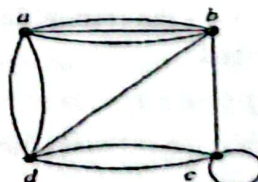
(Answer any 5 (Five) of the following Questions)

1. a) What is Proposition Logic? Which of the following sentences are propositions? Write true or false with the cause: - 2+2
- Birds can fly;
 - Answer the question;
 - $x+2=11$;
 - $5+7=10$;
- b) Define Contraposition and Contradiction. Show that $\neg(P \leftrightarrow Q)$ and $\neg P \leftrightarrow Q$ are logically equivalent. 5
- c) Define Tautology. Show that each of these conditional statements is a tautology by using truth tables. 1+4
- $[\neg p \wedge (p \vee q)] \rightarrow q$
 - $[(p \vee q) \wedge (p \rightarrow r) \wedge (q \rightarrow r)] \rightarrow r$
2. a) Define predicate logic. Let $Q(x, y)$ denote the statement " $x = y + 3$ ". What are the truth values of propositions $Q(1,2)$, $Q(2,3)$, $Q(4,1)$, and $Q(3,0)$? 4
- b) Translate each of these statements into logical expressions using predicates, quantifier- 4
- No one is perfect.
 - Not everyone is perfect.
 - All your friends are perfect.
 - At least one of your friends is perfect.
- c) Define Rules of Inference. Briefly describe it with proper examples. 6

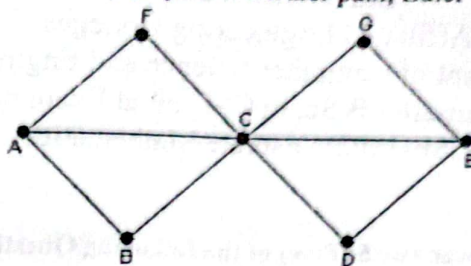
3. a) Define Graph. Use the graph below to answer the questions i – v. 6



- What is the degree of vertex B?
 - What is the degree of vertex H?
 - Is vertex D odd or even?
 - Name at least two vertices that are adjacent to B.
 - Name a path starting at vertex A and ending at vertex F.
- b) What are the differences between simple graph and multi-graph? Draw the adjacency matrix of following graph. 2+2



- c) Determine whether the following graph has a Euler path, Euler circuit, or neither.



4. a) Explain reflexive, antisymmetric and transitive relation with proper example. 6
 b) Define poset and equivalence relation with proper example. 3
 c) Draw the Hasse diagram representing the partial ordering $\{(a, b) | a \text{ divides } b\}$ on $\{1, 2, 3, 4, 6, 8, 12\}$. Find the maximal and minimal elements of the poset $(\{1, 2, 3, 4, 6, 8, 12\}, |)$ 5
5. a) In a survey of 60 people, it was found that 25 read Newsweek magazine, 26 read Time, 26 read Fortune, 9 read both Newsweek and Fortune, 11 read both Newsweek and Time, 8 read both Time and Fortune, 3 read all magazines. 6
 i) Find the number of people who read at least one of the magazines.
 ii) Find the number of people who read exactly one magazine.
 b) Consider Universal set U and two sets A and B . Shade the following sets using Venn diagram. 4
 i) $A \cap B$ ii) $(B/A) \cap C$
 c) Describe Universal and Existential quantifier with example. 4

6. a) Consider the following five relations on the set $A = \{1, 2, 3, 4\}$: 6
 $R = \{(1, 1), (1, 2), (2, 3), (1, 3), (4, 4)\}$
 $S = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 3), (4, 4)\}$
 $T = \{(1, 3), (2, 1)\}$
 $U = A \times A$, the universal relation
 $V = \emptyset$, the empty relation
 Determine whether or not each of the above relations on A is
 i) Reflexive ii) Symmetric iii) Transitive
 b) Let $A = \{1, 2, 3, 4\}$, $B = \{4, 5, 6\}$ and $C = \{6, 7, 8\}$. M is a relation from A to B and N is a relation from B to C , which is given by 4
 $M = \{(x, y) : x + y = 7\}$
 $N = \{(x, y) : y - x = 1\}$

Determine M and N .

- c) What are the Floor and Ceiling Function? Describe with Example. 4
7. a) What is recursion? A function f is defined recursively by $f(0) = 3$, $f(n+1) = 2f(n) + 3$; Find the value of $f(4)$. 4
 b) Find the solution to the recurrence relation: $a_n = 6a_{n-1} - 11a_{n-2} + 6a_{n-3}$ with the initial conditions $a_0 = 2$, $a_1 = 5$ and $a_2 = 15$. 5
 c) What is recurrence relation? Find the recurrence solution for the following recurrence relation using substitution method. 5
 $a_n = a_{n-1} + 3$, $n \geq 2$ and $a_0 = 2$ for $n = 1$.